

GAUGE: Bridging the Command–Motion Gap in Black-Box Quadrupeds

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Abstract—Commercial quadrupeds expose desired planar velocity commands while hiding the controller and dynamics that determine realized motion. Cross-axis coupling, asymmetric response, and deployment shifts can therefore preserve balance while corrupting the variable used by a navigation planner. We present GAUGE (Goal-Aware Uncertainty-Guided Exploration), a task-aware active calibration and shift-recovery layer for opaque quadruped velocity interfaces. GAUGE learns a coupled Bayesian response model and allocates hard-authorized trials according to a planner-facing task distribution. The same posterior supports validated stopping, bounded inverse compensation, and residual-triggered active recovery. On 183 passive Unitree Go2 trials, coupled affine calibration reduces held-out root-mean-square error (RMSE) by 54.5% relative to raw commands. Across controlled synthetic families, task weighting reaches a joint accuracy–uncertainty target in 18.67 trials, saving 39.5% over Latin hypercube sampling and 18.8% over D-optimal design; a no-task ablation and a task-distribution negative control localize the benefit to the declared task measure. Fixed-configuration Isaac Lab experiments yield 9.3–31.7% held-out RMSE reductions. Across four shifts, detection reaches 99.3–100%, recovery reaches 97.9–100%, and task-weighted selection improves early recovery over passive updating. Across six held-out Isaac Sim maps, failure-aware navigation is noninferior to a dense reference. A fixed-planner Go2 comparison further shows more continuous qualitative progress under command compensation. Together, these results show how an opaque velocity interface can become a task-conditioned calibrated control variable.

I. INTRODUCTION

A quadruped can remain upright while executing the wrong motion. High-level controllers accept a desired planar velocity $\mathbf{u} = [v_x, v_y, \omega_z]^\top$, but the steady body velocity may contain gain error, dead zones, saturation, and cross-axis coupling. These errors are particularly consequential when a planner assumes that the exposed command is a calibrated control variable. With a limited interaction budget, calibration must decide which executable commands to measure when deployment occupies only a structured subset of the admissible command space.

Three lines of prior work expose the missing connection. Taouil *et al.* learned center-of-mass and footstep response models from exhaustive command permutations of a black-box quadruped controller and integrated them into footstep planning [1]. Carrillo *et al.* selected calibration configurations by downstream tool-center-point error and used that task-space quantity to assess when calibration was sufficient [2]. SPI-Active optimizes legged-robot excitation using Fisher information, but identifies simulator-facing physical parameters for sim-to-real transfer [3]. This leaves a distinct calibration problem: directing a limited trial budget toward the external command interface itself when neither the low-level controller nor its physical model is available.

Legged adaptation usually changes the controller or estimates variables used inside it. Rapid Motor Adaptation conditions a learned policy on latent environment estimates [4]; safe Bayesian optimization can tune feedback gains directly [5]; and online system identification can estimate mass and center of mass before updating a model-based controller [6]. Residual and embodiment adaptation likewise operate on controller or policy state [7], [8], [9]. GAUGE instead calibrates the user-visible command-to-motion interface while leaving the hidden dynamics and controller unchanged.

GAUGE treats calibration as a deployment lifecycle (Fig. 1). A planner-facing task measure specifies which desired velocities the planner is likely to request. The response posterior then selects informative commands, quantifies whether validation is sufficient, and supplies a bounded inverse from desired motion to the opaque interface. A non-learned authorization layer remains the sole arbiter of execution. Frozen residual signatures detect when the calibrated map has changed; a latched alarm reopens task-weighted calibration and returns to validation before normal operation resumes.

The contributions are threefold. First, we formulate opaque velocity-interface calibration as deployment-task-conditioned Bayesian experimental design, with planner-facing predictive uncertainty determining trial selection. Second, we use one response posterior throughout an operational lifecycle that separates information ranking from hard authorization and supports validated stopping, bounded inversion, and residual-triggered active recovery. Third, we establish an evidence ladder: passive Go2 data test model need; controlled synthetic and Isaac experiments isolate acquisition and recovery mechanisms; held-out navigation tests downstream consequence; and a fixed-planner physical comparison provides qualitative interface-level evidence.

II. RELATED WORK

A. Black-Box Command Models

Velocity-controlled robots can be calibrated from externally measured motion without access to their internal dynamics. Li and Stücker jointly calibrate a velocity-control kinematic model within visual–inertial odometry [10]. For quadrupeds, Taouil *et al.* learn motion models of an opaque controller and use them in an A*-based footstep planner [1]. Their calibration data exhaust the command permutations of interest. GAUGE treats data collection itself as the limited resource and allocates it according to the command distribution induced by the downstream planner.

The predictive mean is $\mu_a(\mathbf{u}) = \mathbf{m}_a^\top \phi(\mathbf{u})$. The response variance used by bounded inversion is $s_a^2(\mathbf{u}) = \phi(\mathbf{u})^\top \Sigma_a \phi(\mathbf{u}) + \sigma_a^2$; coverage of a future measured response also adds its observation variance r_a . Acquisition uses only the reducible epistemic term. Coupled input features represent cross-axis response while the three output posteriors remain conditionally independent.

C. Task-Weighted Active Calibration

Uniformly reducing uncertainty over the admissible envelope wastes trials when deployment concentrates on a structured command distribution. For candidate $\mathbf{c} \in \mathcal{C}$, task-weighted integrated variance reduction (IVR), obtained by a Sherman–Morrisson update, gives the conditional one-step reduction in Eq. (1):

$$\mathcal{I}(\mathbf{c}) = \sum_{a=1}^3 \sum_{j=1}^G w_j \frac{(\phi(\mathbf{g}_j)^\top \Sigma_a \phi(\mathbf{c}))^2}{\sigma_a^2 + \tilde{r}_a(\mathbf{c}) + \phi(\mathbf{c})^\top \Sigma_a \phi(\mathbf{c})}. \quad (5)$$

The optional $\tilde{r}_a(\mathbf{c})$ predicts candidate observation-noise variance; the posterior update always uses the realized r_a . The selector maximizes $\mathcal{I}(\mathbf{c}) - \lambda_m P_{\text{mag}}(\mathbf{c}) - \lambda_\Delta P_{\text{move}}(\mathbf{c})$, where the two penalties limit command magnitude and travel. Greedy batches use outcome-independent covariance updates. This score ranks information value; the hard authorization layer makes the execution decision.

D. Authorization, Stopping, and Bounded Inversion

The hard filter checks state validity, localization, battery, attitude, base height, command and slew bounds, combined normalized linear–angular load, and workspace occupancy. It fails closed if no ranked candidate remains, and an independent 50 Hz monitor can zero commands. Statistical value cannot relax an execution limit.

Stopping requires minimum coverage and three consecutive checks of held-out validation RMSE and task-facing uncertainty; hard budgets take priority. During deployment, the inverse searches the bounded candidate set for planner target $\mathbf{v}^*$ and minimizes

$$\|W(\mu(\mathbf{u}) - \mathbf{v}^*)\|_2^2 + \lambda \|W(\mathbf{u} - \mathbf{v}^*)\|_2^2 + \rho \sum_a s_a^2(\mathbf{u}), \quad (6)$$

where W normalizes the three axes and $\lambda, \rho \geq 0$ trade command regularization against predictive risk. The selected command then passes through the hard filter.

E. Shift Detection and Active Recovery

Shift monitoring cycles through four authorized commands $(-.20, 0, -.20)$, $(-.20, 0, .20)$, $(.12, -.10, -.15)$, and $(.12, .10, .15)$. Commissioning freezes the corresponding residual signatures $\mathbf{r}_j^0 = \mathbf{y}_j - \mu(\mathbf{u}_j)$. For a later response to signature command $j(t)$, the current residual is $\mathbf{r}_t = \mathbf{y}_t - \mu(\mathbf{u}_{j(t)})$, and the detector evaluates

$$d_t = \left\| \text{diag}(.14, .08, .18)^{-1} (\mathbf{r}_t - \mathbf{r}_{j(t)}^0) \right\|_2. \quad (7)$$

An alarm requires $d_t > .30$ on at least two of the last four valid monitor trials and a two-trial minimum dwell. The

alarm latches, inflates epistemic covariance by a factor of eight without changing the posterior mean, and opens a 12-command task-weighted recovery budget. Normal validation stopping resumes after recovery.

IV. EXPERIMENTS AND RESULTS

We organize the evaluation around four questions. **Q1:** Is coupled calibration necessary for a real black-box quadruped velocity interface? **Q2:** Does task weighting reduce calibration effort relative to passive, parameter-oriented, and task-agnostic designs? **Q3:** Can the calibrated interface detect deployment shifts and recover without conflating information ranking with execution authorization? **Q4:** Do command-level gains translate into navigation? Hardware evidence comprises passive model data and a qualitative fixed-planner comparison; active-selection and recovery mechanisms are tested in controlled synthetic and fixed-configuration Isaac experiments.

A. Coupled Response Modeling Is Necessary

The passive Go2 data show that an intercept alone cannot absorb off-axis response (Fig. 2a). The dataset contains 183 valid trials in three 61-trial sessions, measured with a Livox MID-360 running FAST-LIO [16]. Leave-one-session-out folds compare raw commands with diagonal and coupled linear/affine maps. Pooled RMSE is 0.06605 for raw commands, 0.04828 for diagonal linear, 0.04563 for diagonal affine, 0.03517 for coupled linear, and 0.03009 for coupled affine. Coupled linear therefore improves on diagonal affine without adding an intercept. The coupled-affine map reduces error by 54.45% relative to raw commands and 34.07% relative to diagonal affine; its three held-out-session RMSE values are 0.02888, 0.03158, and 0.02974. Lateral RMSE falls from 0.08459 to 0.02569, while the yaw-to-lateral coefficient remains 0.119–0.129 across holdouts. A 30-trial fit is within 3.33% of the 122-trial dense fit.

Controlled closed-loop runs test whether calibration improves realized motion under physics. Four Isaac Lab contexts [17] use 20 paired seeds, 12 calibration commands, and eight held-out commands with a common robust constant-twist estimator. Raw-to-calibrated RMSE is 0.15199→0.10635 for affine distortion, 0.15054→0.10282 for dead-zone distortion, 0.10741→0.09742 for low friction with load and center-of-mass offset, and 0.09554→0.07849 on rough terrain with load and offset (Fig. 2b). Relative reductions span 9.30–31.70%, and all 20 paired seeds improve in every context. The hardware inference is passive model need; the paired Isaac runs supply the closed-loop intervention test.

B. Task Weighting Reduces Calibration Effort

The acquisition study uses 20 paired seeds in affine, dead-zone, and heteroscedastic synthetic families. All methods share a six-command seed and a 160-trial horizon over $[-1, 1] \times [-.5, .5] \times [-1.5, 1.5]$, with $\|(v_x, v_y)\|_2 \leq 1$. Frozen draws vary gain, off-axis coupling, bias, saturation, and noise. The declared task measure weights forward, left-turn,

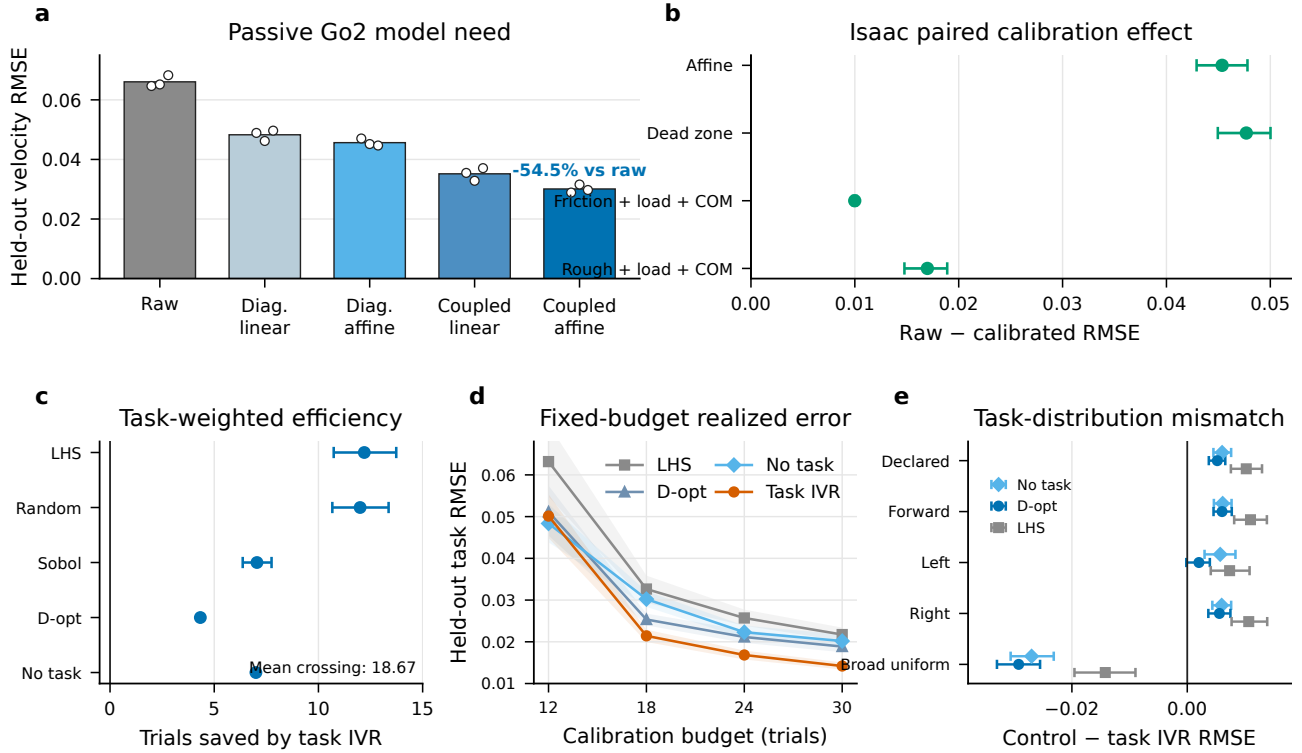


Fig. 2. Why coupled, task-directed calibration is needed. (a) Passive Go2 held-out RMSE across nested response models. (b) Seed-paired raw-minus-calibrated RMSE in four Isaac Lab contexts. (c) Trials saved by task IVR relative to five controls. (d) Realized held-out task RMSE at fixed budgets. (e) Frozen acquisitions evaluated under task-distribution reweighting; positive values favor task IVR over each control, while the broad-uniform negative control reverses the ordering. Error bars and bands are paired 95% bootstrap intervals.

and right-turn command components by $(.50, .25, .25)$. The stopping target is $\text{RMSE} \leq 0.04$ and integrated epistemic variance ≤ 0.0015 . Random, Latin hypercube sampling (LHS), Sobol, D-optimal, no-task IVR, and a dense oracle form the controls; acquisition penalties are zero. A hidden 400-command grid defines crossing and a disjoint 1,024-command grid audits it. Paired Wilcoxon contrasts use Holm correction, and paired bootstrap intervals use 4,000–5,000 resamples.

Task IVR reaches the joint target in 18.67 trials, compared with 30.87 for LHS, 30.67 for random, 25.72 for Sobol, 23.00 for D-optimal, and 25.67 for no-task IVR (Fig. 2c). The corresponding savings are 39.52%, 39.13%, 27.41%, 18.84%, and 27.27%; all paired contrasts remain significant after Holm correction. The primary LHS contrast saves 12.20 trials [10.73, 13.73]. The no-task comparison identifies the planner-facing measure, beyond generic active selection, as the source of the gain.

Realized held-out error preserves this ordering (Fig. 2d). At budgets 18, 24, and 30, task IVR RMSE is 0.02143, 0.01684, and 0.01420. Its paired improvement is 0.00392–0.00466 over D-optimal and 0.00545–0.00881 over no-task IVR; all six 95% intervals exclude zero. The uncertainty gate binds in all 60 active conditions, with family-level crossings at 18, 18, and 20 trials. The disjoint audit yields crossing RMSE 0.01796–0.02315 and axiswise predictive

coverage 0.917–0.955. Supplying exact heteroscedastic candidate noise moves one crossing from 20 to 21 trials and preserves the 11-trial lead over the two principal controls.

The benefit follows the declared deployment support. Frozen acquisitions retain positive paired differences relative to task IVR under forward- and right-heavy navigation mixtures, and the left-heavy D-optimal contrast becomes positive by 30 trials (Fig. 2e). When evaluation expands to the uniform authorized envelope, the ordering reverses at 24 trials and task IVR error becomes 3.21 times its error under the declared measure. Task-directed calibration thus trades global coverage for lower error on the planner-facing distribution.

C. Operational Robustness and Shift Recovery

Replay tests first verify the separation between ranking and authorization. Across 60 acquisition traces, the hard filter rejects 300/300 hazardous proposals, rejects 0/20 authorized controls, and catches all 160 injected state faults. The stopping rule makes no premature decision on 60 trajectories; median and p95 overshoot are two trials. The 50 Hz physics monitor has a maximum 20 ms abort interval. These tests establish the implemented authorization and stopping behavior under replay and simulation, not a physical safety guarantee.

Four held-out shifts are evaluated: combined friction, load, and gain; gain and coupling; physical and nonlinear response; and load, center of mass, and gain (Fig. 3a). Each shift pools

two disjoint 72-seed blocks. Fixed-model, passive-update, and GAUGE arms share 12 initial commands, four residual signatures, five monitor commands, a 12-command recovery budget, and rolling validation. The pre-specified early endpoint averages recovery trials 4–9. Stationary monitoring covers 120 sequences and 12,000 valid trials.

The stationary monitor produces 0/120 sequence alarms, with a two-sided exact 95% upper bound of 0.0303. Detection is 143/144 for two shifts and 144/144 for the other two; the lowest exact interval is [0.9619, 0.9998]. GAUGE recovers in 141/144 friction/load/gain trials and 144/144 elsewhere, with lowest interval [0.9403, 0.9957]. Worst p95 detection and recovery delays are 3.9 and 9 trials, equivalent to 10.1 s of monitor commands and 46.8 s for the fixed recovery-plus-validation schedule.

Active recovery lowers early RMSE relative to passive updating by 0.00936, 0.01053, 0.01847, and 0.01192 across the four shifts; every paired 95% interval is positive (Fig. 3b). Terminal RMSE ranges from 0.09762 to 0.11227, and every interval upper endpoint remains below 0.116 and the declared 0.14 gate (Fig. 3c). The selector ablation keeps detection, updating, and recovery budgets fixed. Task IVR lowers early RMSE by 0.02192 versus D-optimal, 0.01583 versus no-task IVR, 0.00937 versus LHS, and 0.00986 versus random selection, with positive paired intervals throughout (Fig. 3d). The early-recovery gain therefore depends on the task-directed selector rather than detection and posterior updating alone.

D. Navigation Consequence

Held-out navigation crosses six disjoint maps, seven calibration methods, and 72 paired seeds per map, for 3,024 episodes (Fig. 4). All arms share the controller, 50 Hz interlock, 10 Hz waypoint follower, geometry, and randomization. GAUGE and matched controls use 12 calibration commands; the dense reference uses 30, followed by eight common validation commands. Task IVR ranks a 512-command design pool under the planner-facing task measure. Routes span 2.85–3.60 m, with a 0.79 m minimum opening and 0.20 m collision radius. Success requires reaching the 0.25 m goal ball without collision or simulator termination (a serious event) by 60 s, and every failure retains the 60-s cap. The declared noninferiority margins are a 1.25 capped-time ratio and a -0.05 success difference.

GAUGE succeeds in 72/72 episodes on five maps and 70/72 on weighted arc; raw commands succeed in 0/72 or 1/72 (Fig. 4a). Every GAUGE collision count is zero. The 95% lower endpoints of the paired raw-minus-GAUGE capped-time gain are 23.45–32.68 s (Fig. 4b). At the same 12-trial budget, task IVR reduces validation RMSE over D-optimal by 9.98–12.28% on every map; its advantage over no-task IVR is positive on five maps and unresolved on S-bend (Fig. 4c).

The worst GAUGE/dense capped-time ratio upper bound is 1.0736, below the 1.25 margin, and every paired success-difference lower bound exceeds -0.05 (Fig. 4d). The two weighted-arc timeouts are neither collisions nor simulator

terminations. Acquisition takes 31.2 s for GAUGE and 78.0 s for dense calibration, or 52.0 and 98.8 s after common validation. Across its 432 episodes, GAUGE records 10 authorization interventions and no simulator termination.

A qualitative physical comparison uses the same fixed DRL-DCLP local planner [18] with either direct commands or the frozen GAUGE inverse. One complete run per condition is recorded from a fixed camera in the same obstacle layout. At matched wall-clock times, the direct run accumulates poses beside the central obstacle and visibly contacts it, whereas the compensated run progresses more continuously through the layout (Fig. 1, bottom). The fixed software path supports an interface-level behavioral comparison; one run per condition does not estimate a trajectory-level effect or run-to-run variability.

V. DISCUSSION AND LIMITATIONS

Task weighting converts deployment structure into calibration efficiency. When the planner concentrates commands on a structured subset of the interface, global identification spends trials on responses that do not affect the task. The no-task ablation shows that sequential information gain alone does not achieve the same trial efficiency, and the broad-uniform negative control gives the complementary boundary: once deployment support expands, the advantage disappears and can reverse. The task measure is therefore part of the calibration specification, not a fixed property of the robot.

Each evidence tier answers a different part of the argument. Passive Go2 data establish the need for a coupled external response model. Synthetic controls isolate the acquisition principle, and fixed-configuration Isaac experiments test closed-loop calibration, authorization logic, shift detection, and active recovery. Held-out simulator maps connect command calibration to quantitative navigation outcomes. The fixed-planner Go2 comparison adds qualitative behavior-level evidence without replacing a replicated physical endpoint.

The current model targets a steady command–response map. Strong history dependence, rapid transients, or gait and controller switching require an expanded response state or separate models. A changed planner distribution requires a new task measure, as the support-expansion control demonstrates. Quantitative physical evaluation of online active acquisition, structured shifts, and recovery remains the main step needed to test whether the complete lifecycle transfers beyond the fixed simulator configuration.

VI. CONCLUSION

GAUGE turns an opaque quadruped velocity interface into a task-conditioned calibrated control variable. It directs a limited trial budget toward planner-facing uncertainty, validates the learned response before bounded compensation, and reopens active calibration when residual signatures indicate a shift. Passive Go2 data establish coupled model need; controlled synthetic and Isaac experiments establish task-weighted acquisition and recovery; and held-out navigation establishes downstream consequence.

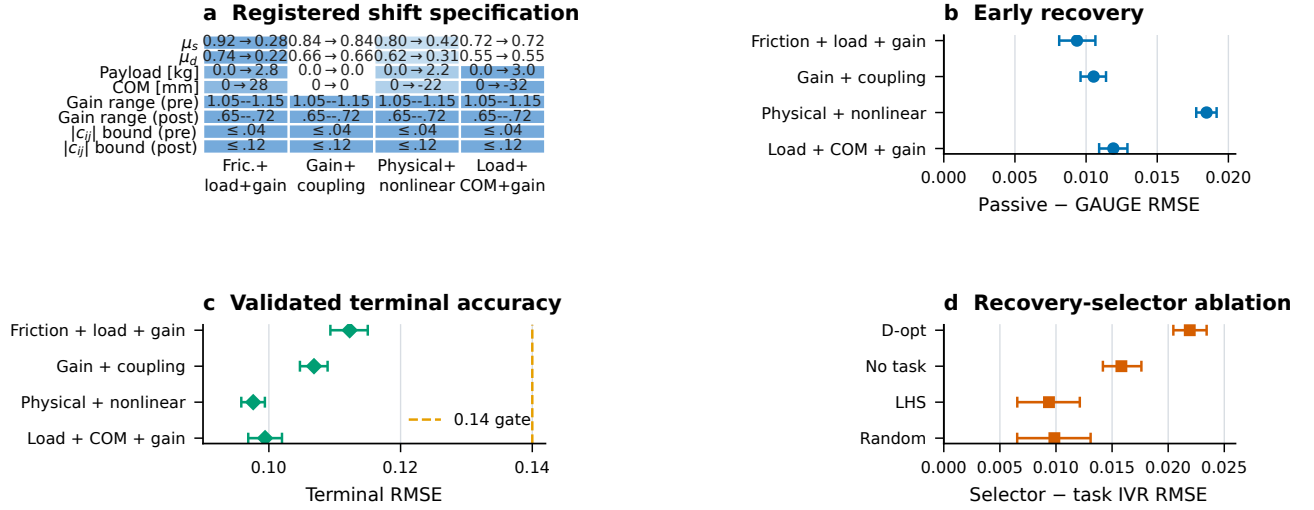


Fig. 3. Operational shift and recovery. (a) Registered pre→post friction, payload, fore-aft center-of-mass, gain, and coupling specification; the third, mixed physical/nonlinear context also introduces dead zones, saturation, lag, and observation noise. Cell shading is normalized within each parameter row. (b) Early RMSE improvement over passive updating. (c) Terminal RMSE against the 0.14 absolute gate. (d) Recovery-selector ablation, averaging the four contexts within each of 30 paired seeds; positive values favor task IVR. Error bars are 95% bootstrap intervals. Panels (b) and (c) pool two disjoint 72-seed blocks per shift; panels (b) and (d) use seed-paired contrasts.

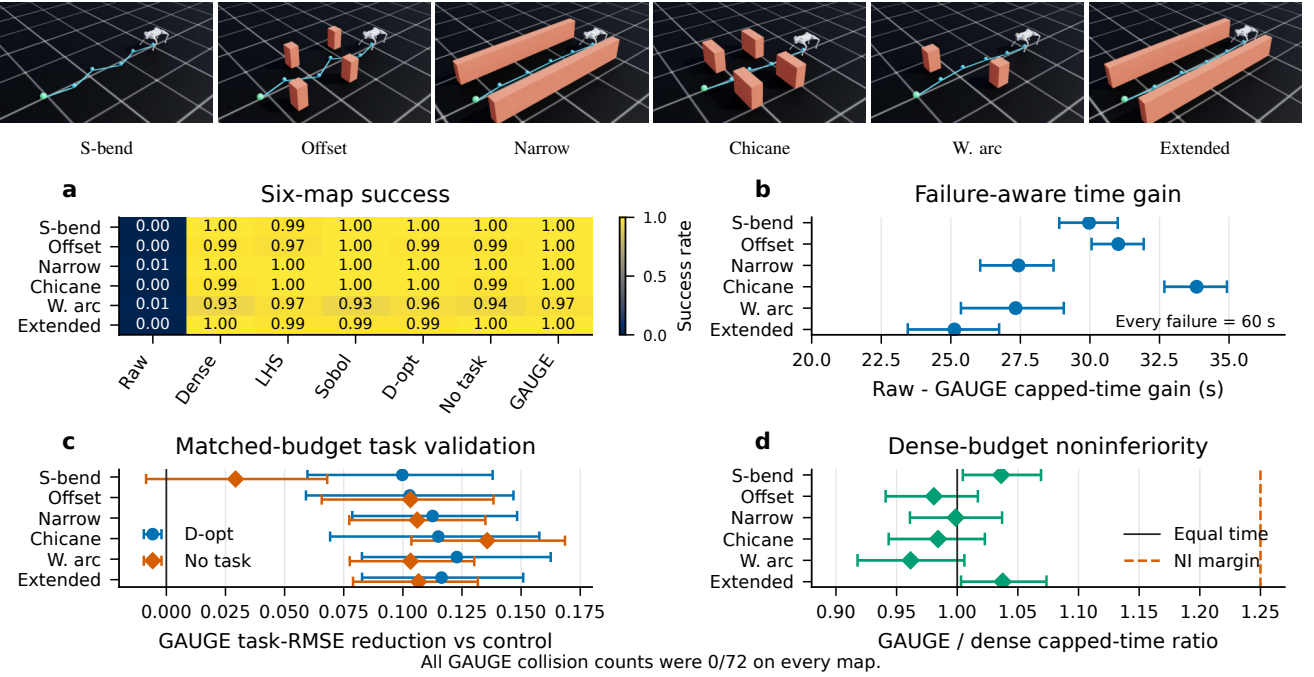


Fig. 4. Navigation consequence on six held-out Isaac Sim maps (72 seeds/cell). Top: map geometry and waypoint overlays. Bottom: (a) success rate; (b) paired raw-minus-GAUGE gain in the failure-aware 60-s capped-time estimand; (c) matched-budget validation-RMSE reduction relative to D-optimal and no-task IVR; and (d) GAUGE/dense capped-time ratio with the 1.25 noninferiority margin. Every GAUGE collision count is 0/72.

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